



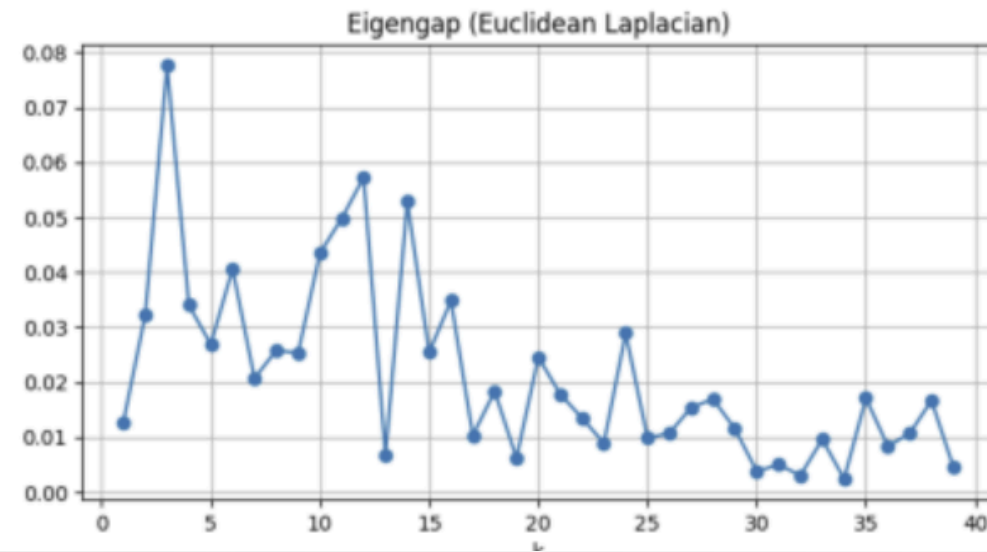
POP PYRAMID UPDATES

TASK EXPLANATION

- The task was to cluster countries based on population pyramids to understand how population structures change over time.
- It was explained to do this using a tensor-based approach — combining Country × Age Group × Year data into one large matrix.
- We were told to analyze demographic patterns and compare how countries' populations evolve.
- For example, it was shown how China's population pyramid changes over time — from expansive to stationary, and later to constrictive — and we had to do this type of analysis for all selected countries.
- I used data for 177 countries, 21 age groups, and 1950–2025 years.
- Applied Euclidean Laplacian spectral and hierarchical clustering, and found 3 clear clusters:
- Expansive (Developing)
- Stationary (Transitional)
- Constrictive (Developed)
- This shows how countries move through different demographic stages as their population shapes evolve over time.

RESULTS

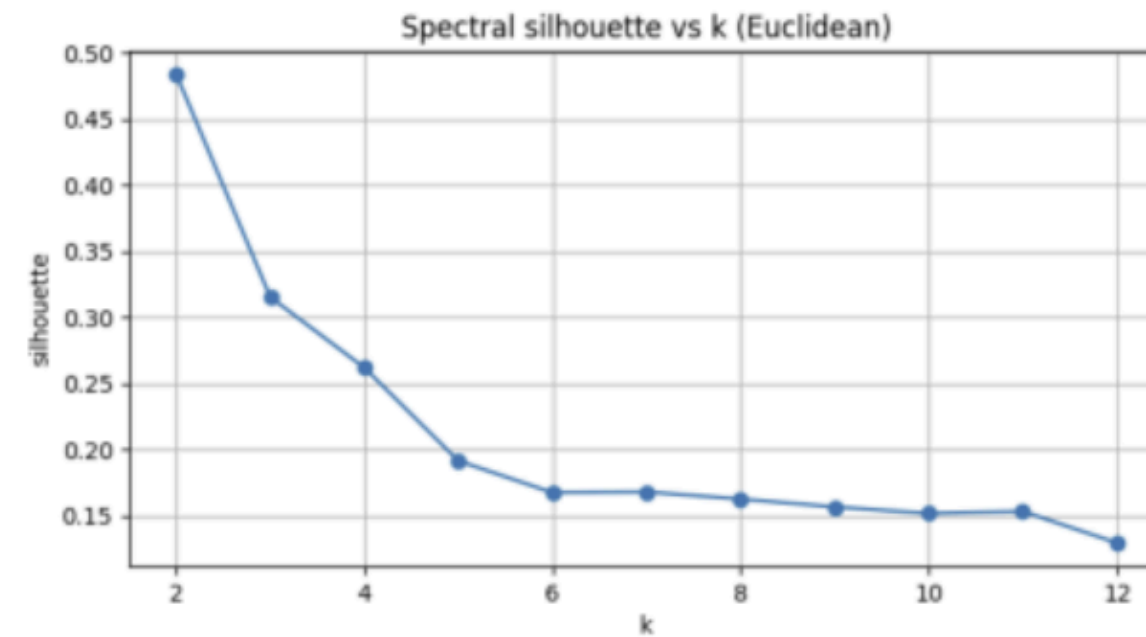
Eigengap (Optimal $k \approx 3$)



Eigengap Plot (Optimal $k \approx 3$)

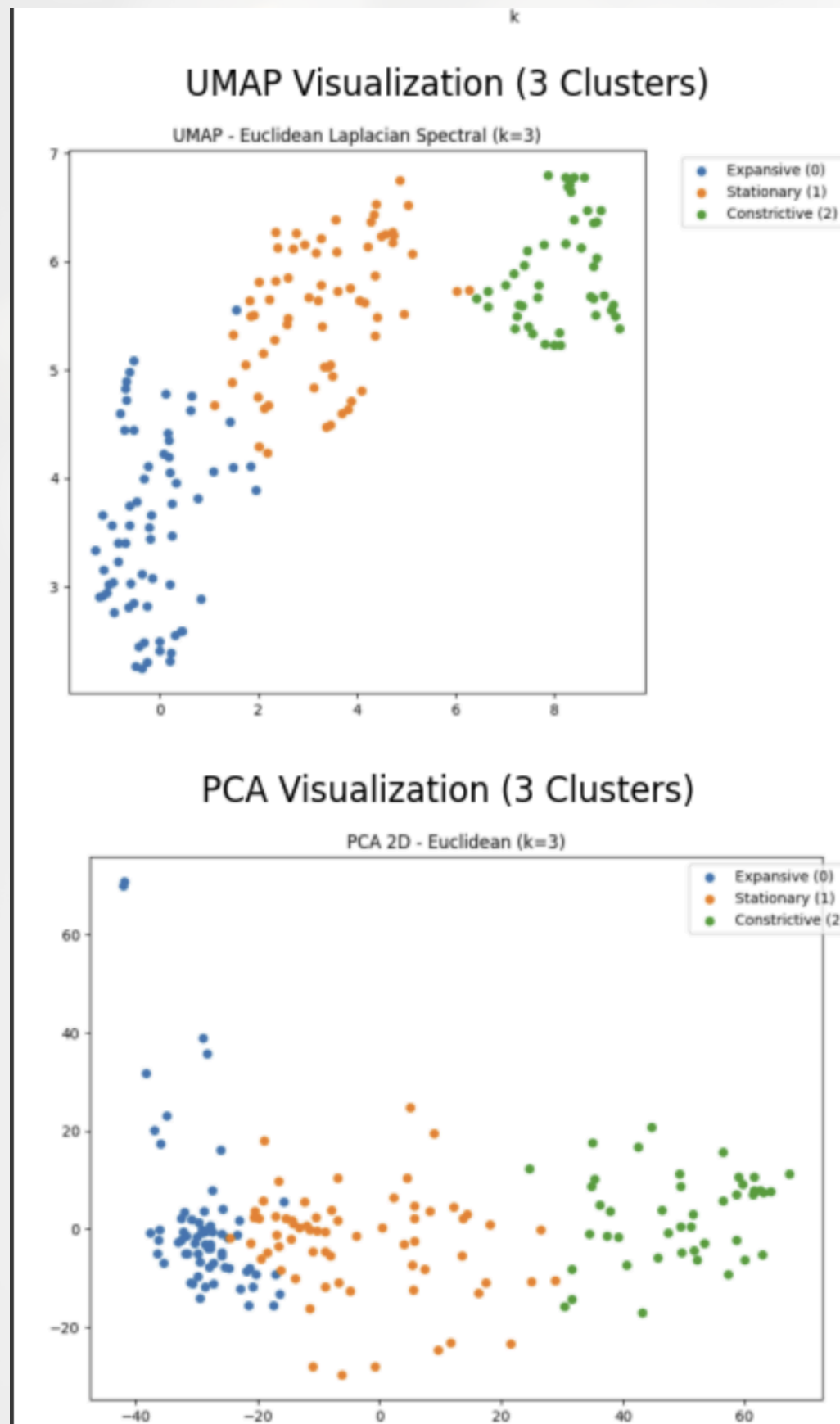
- The eigengap shows where the largest jump occurs between consecutive eigenvalues of the Laplacian matrix.
- Here, the biggest gap is around $k = 3$, which means the data naturally forms three main groups.
- These correspond to demographic stages (Expansive, Stationary, Constrictive).

Silhouette Scores vs k



- The silhouette score measures how well countries fit within their clusters.
- The highest score appears near $k = 3$, confirming that three clusters provide the most stable and meaningful grouping.

RESULTS

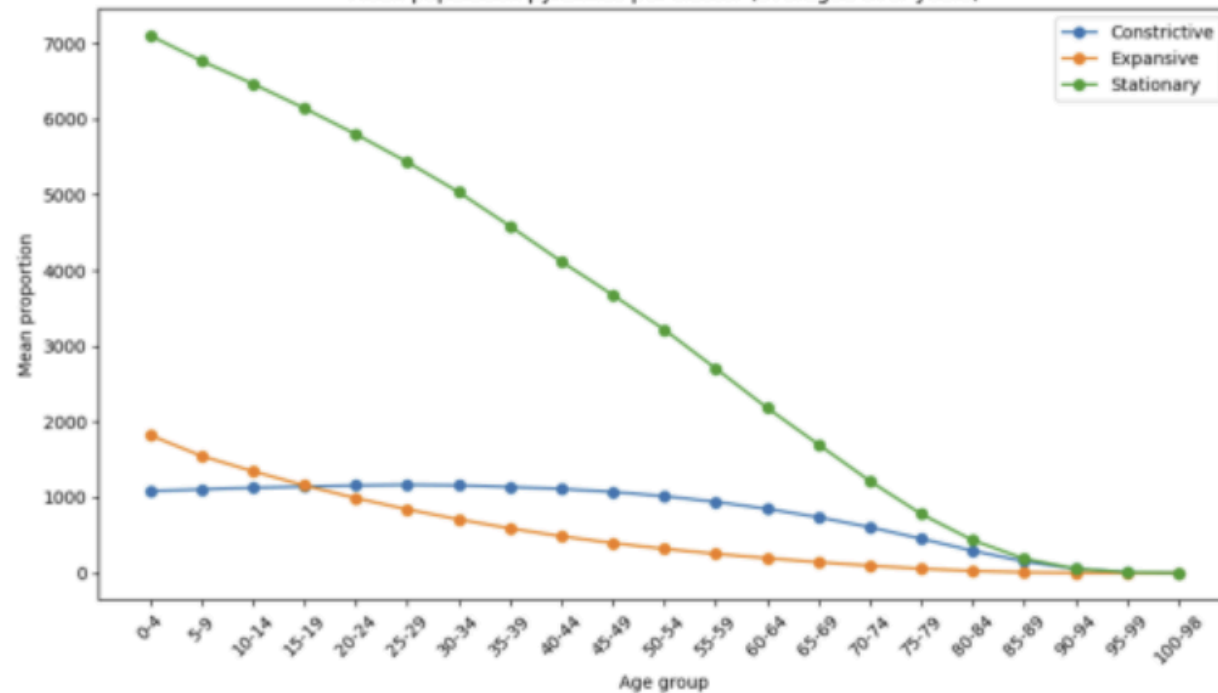


- UMAP revealed clear separation among Expansive, Stationary, and Constrictive clusters, preserving local structure and highlighting subtle demographic differences
- PCA confirmed the same clusters using linear projection, showing global variance and supporting the overall clustering logic.
- **Both methods consistently grouped countries by age distribution patterns, validating the demographic categories across techniques.**

RESULTS

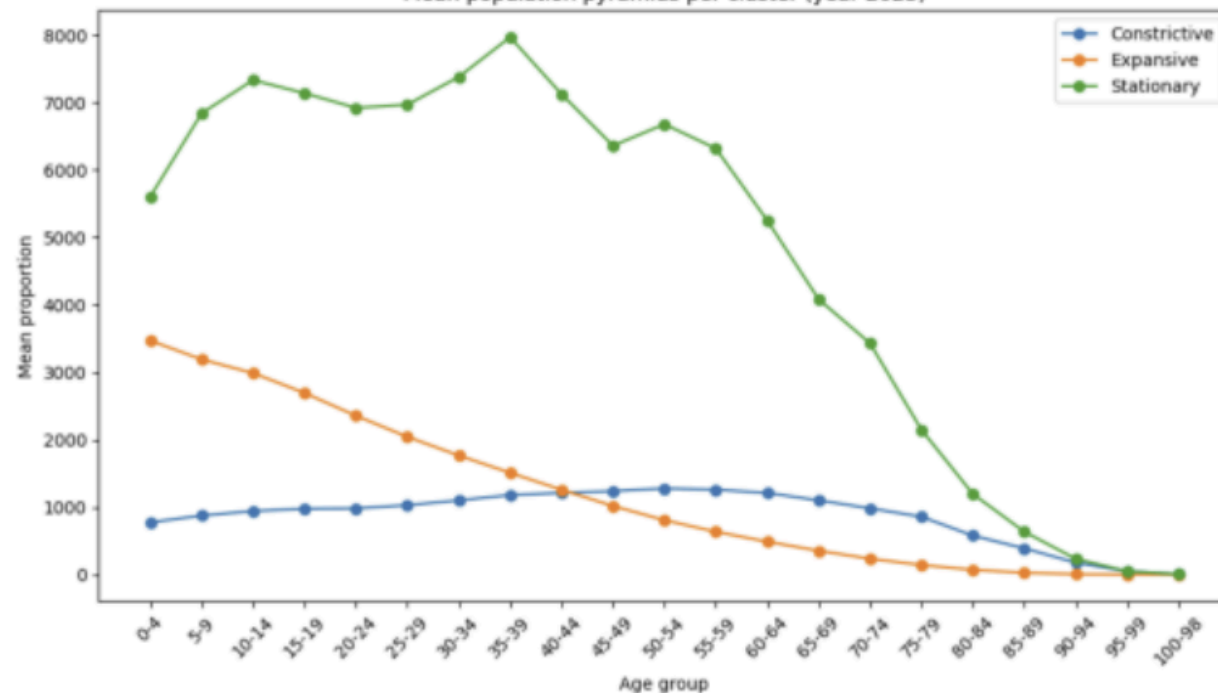
Mean Population Pyramids (All Years)

Mean population pyramids per cluster (averaged over years)



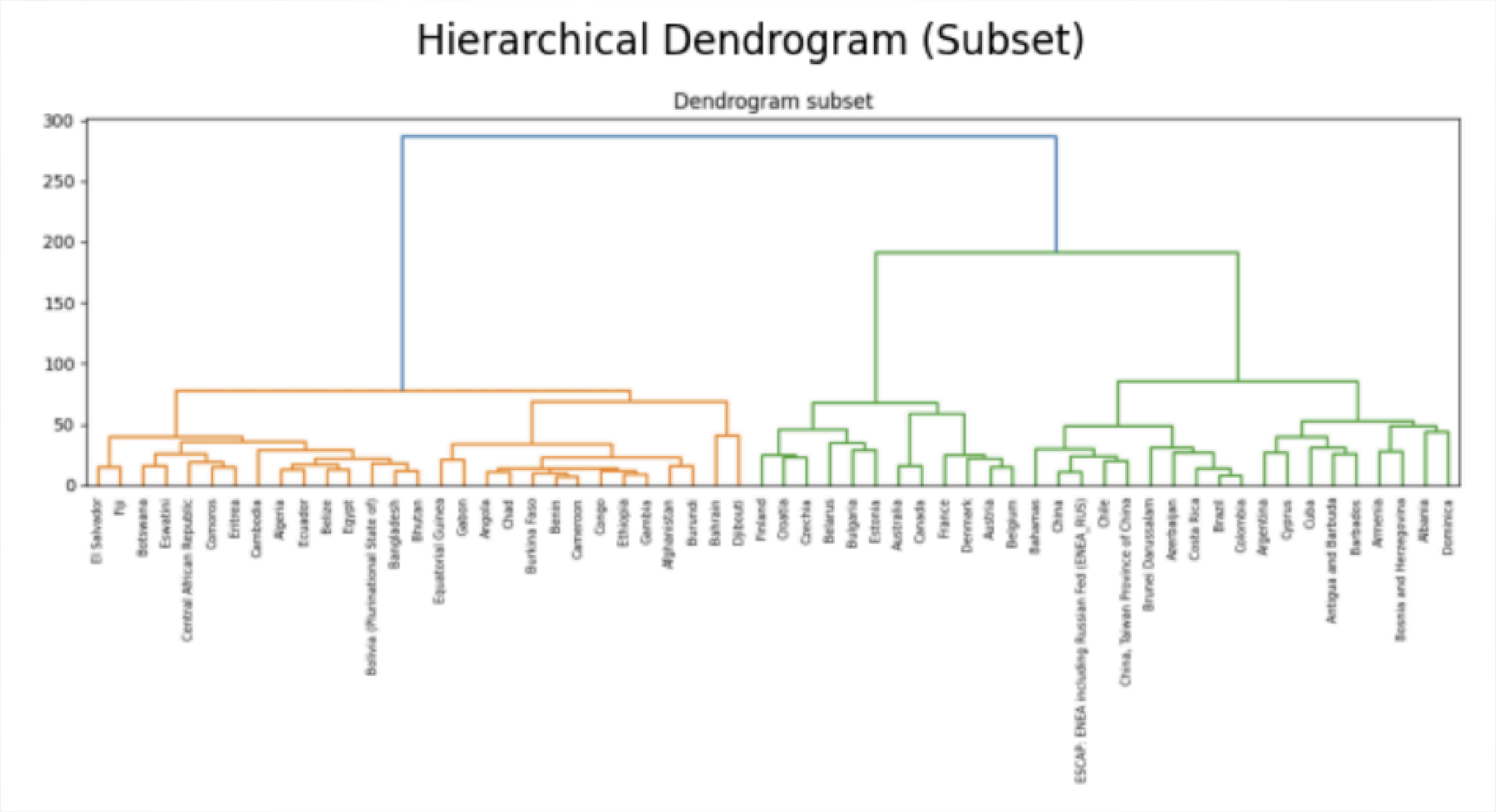
Mean Population Pyramids (2025)

Mean population pyramids per cluster (year 2025)



- The Mean Population Pyramids (All Years) graph shows consistent demographic patterns across clusters. Expansive countries have a wide base, indicating high birth rates and a young population. Stationary countries maintain a relatively stable distribution across age groups, while Constrictive countries show a shrinking youth population and a growing elderly segment.
- In the 2025 projection, these trends become more pronounced. Expansive populations continue to decline steadily with age, Stationary populations remain balanced but begin to taper in older groups, and Constrictive populations peak in mid-age before declining sharply—highlighting aging societies and future dependency challenges.

RESULTS



RESULTS

- Spectral Cluster 0: Expansive (high young population, e.g., Afghanistan, Angola).
 - Spectral Cluster 1: Stationary (balanced population, e.g., Albania, Algeria).
 - Spectral Cluster 2: Constrictive (older population, e.g., Australia, Belgium).
-
- Hierarchical Cluster 0: Likely Expansive, as it includes countries like Afghanistan, Algeria, and India, which have high young populations (86 countries).
 - Hierarchical Cluster 1: Likely Constrictive, as it includes developed countries like Australia, Belgium, and Canada (43 countries).
 - Hierarchical Cluster 2: Likely Stationary, as it includes countries like Albania, Brazil, and China, which are often in transition (48 countries).

RESULTS

Cluster ID	Label	Description	Example Countries
0	Expansive	High young population (≤14)	Afghanistan, Angola
1	Stationary	Balanced age distribution	Albania, Algeria
2	Constrictive	High old population (≥65)	Australia, Austria

Spectral clustering

Cluster ID	Label	Description	Example Countries
0	Expansive	High young population (≤14)	Afghanistan, Algeria
1	Constrictive	High old population (≥65)	Australia, Austria
2	Stationary	Balanced age distribution	Albania, Antigua and Barbuda

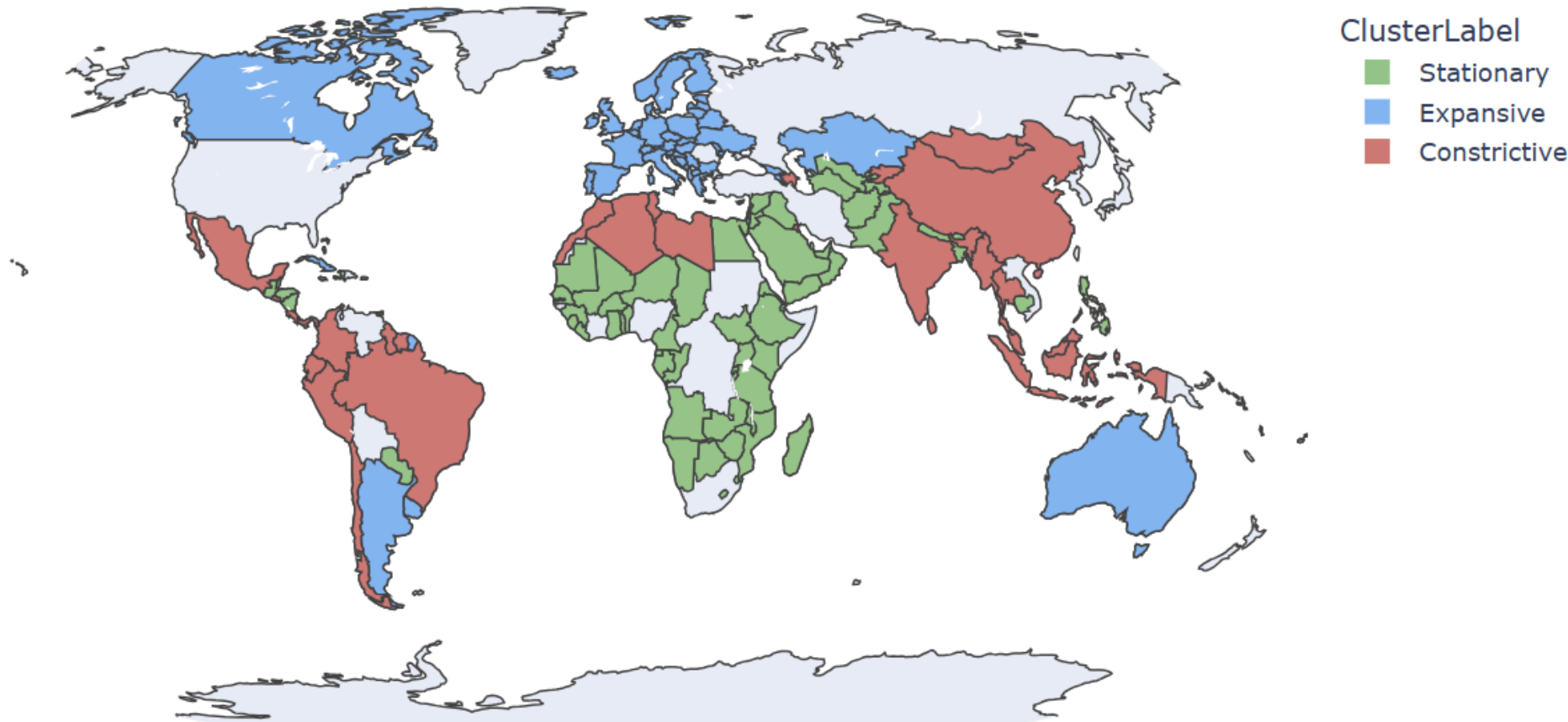
Hierarchical clustering

DYNAMIC CLUSTERING AND EVOLUTION OF POPULATION PYRAMIDS (1950–2025)

- Spectral, Hierarchical, and DTW (Dynamic Time Warping) methods.
- Map clusters to demographic transition stages and dividend potential.
- Visualize shifts in structure (Expansive → Stationary → Constrictive).

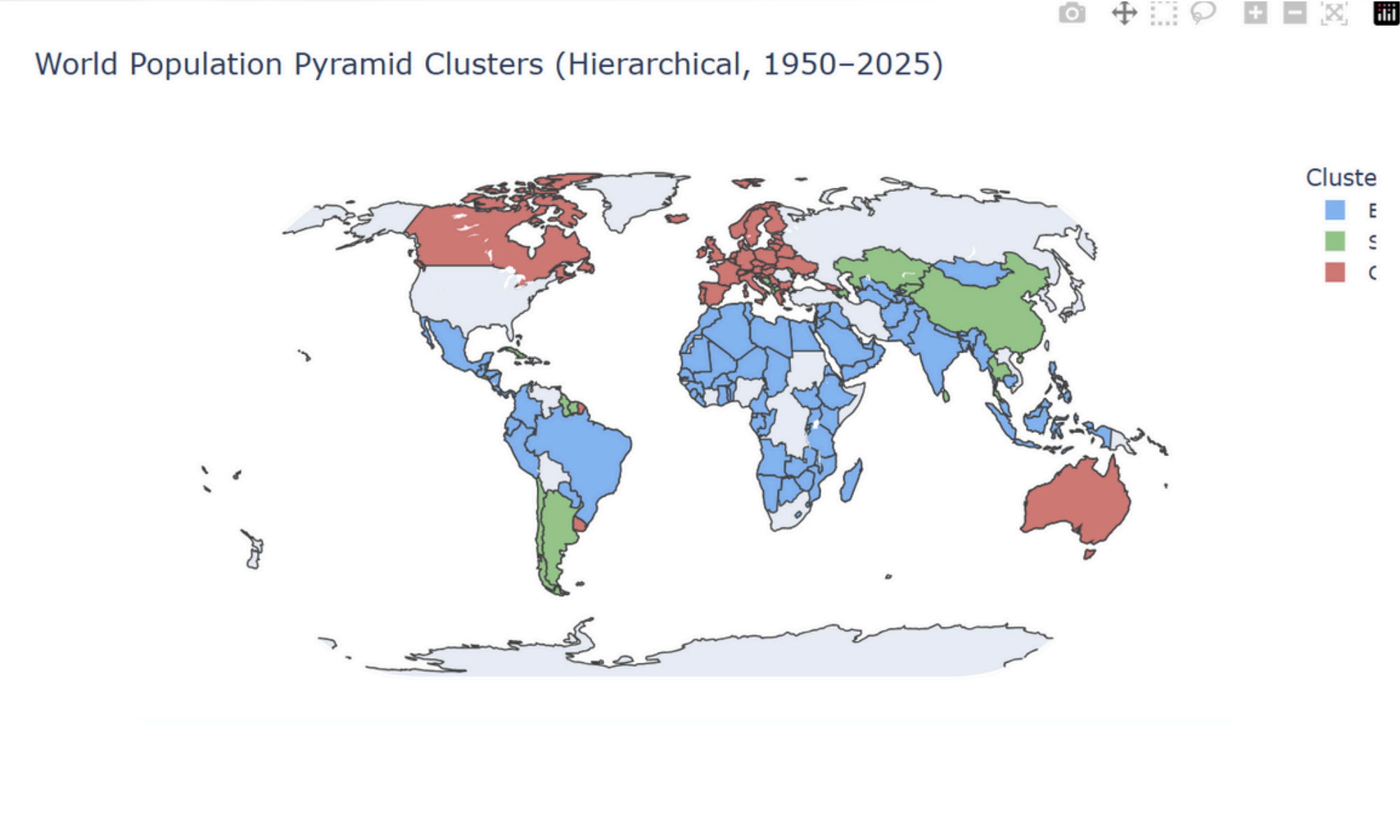
SPECTRAL CLUSTERING RESULTS

World Population Pyramid Clusters (Spectral, 1950–2025)



ClusterType	Countries
Constrictive	Algeria, American Samoa, Antigua and Barbuda, Azerbaijan, Bahamas, Bahrain, Brazil, Brunei Darussalam, Chile, China, China, Taiwan Province of China, Colombia, Costa Rica, Dominica, ESCAP: ENEA including Russian Fed (ENEA_RUS), Ecuador, El Salvador, Fiji, Grenada, Guam, Guyana, India, Indonesia, Iran (Islamic Republic of), Jamaica, Kuwait, Kyrgyzstan, Lebanon, Libya, Malaysia, Mauritius, Mexico, Mongolia, Morocco, Myanmar, Panama, Peru, Saint Lucia, Saint Vincent and the Grenadines, Seychelles, Singapore, Sri Lanka, Suriname, Thailand, Tonga, Trinidad and Tobago, Tunisia, United States Virgin Islands, Venezuela (Bolivarian Republic of)
Expansive	Albania, Argentina, Armenia, Australia, Australia/New Zealand, Austria, Barbados, Belarus, Belgium, Bosnia and Herzegovina, Bulgaria, Canada, Croatia, Cuba, Cyprus, Czechia, Denmark, Estonia, Finland, France, Georgia, Germany, Greece, Hungary, IPCC AR6: Australia, Japan, and New Zealand, Iceland, Ireland, Israel, Italy, Kazakhstan, Latvia, Lithuania, Luxembourg, Malta, Montenegro, Netherlands, North Macedonia, Norway, Poland, Portugal, Puerto Rico, Republic of Moldova, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Ukraine, United Kingdom, Uruguay
Stationary	Afghanistan, Africa, Angola, Bangladesh, Belize, Benin, Bhutan, Bolivia (Plurinational State of), Botswana, Burkina Faso, Burundi, Cambodia, Cameroon, Chad, Comoros, Congo, Djibouti, Egypt, Equatorial Guinea, Eritrea, Eswatini, Ethiopia, Gabon, Gambia, Ghana, Guatemala, Guinea, Haiti, Honduras, Iraq, Jordan, Kenya, Kiribati, Lesotho, Liberia, Madagascar, Malawi, Maldives, Mali, Mauritania, Mozambique, Namibia, Nepal, Nicaragua, Niger, Oman, Pakistan, Paraguay, Philippines, Qatar, Rwanda, Sao Tome and Principe, Saudi Arabia, Senegal, Sierra Leone, Solomon Islands, South Sudan, Syrian Arab Republic, Tajikistan, Timor-Leste, Togo, Turkmenistan, Uganda, United Arab Emirates, United Republic of Tanzania, Uzbekistan, Vanuatu, Yemen, Zambia, Zimbabwe

HIERARCHICAL CLUSTERING RESULTS



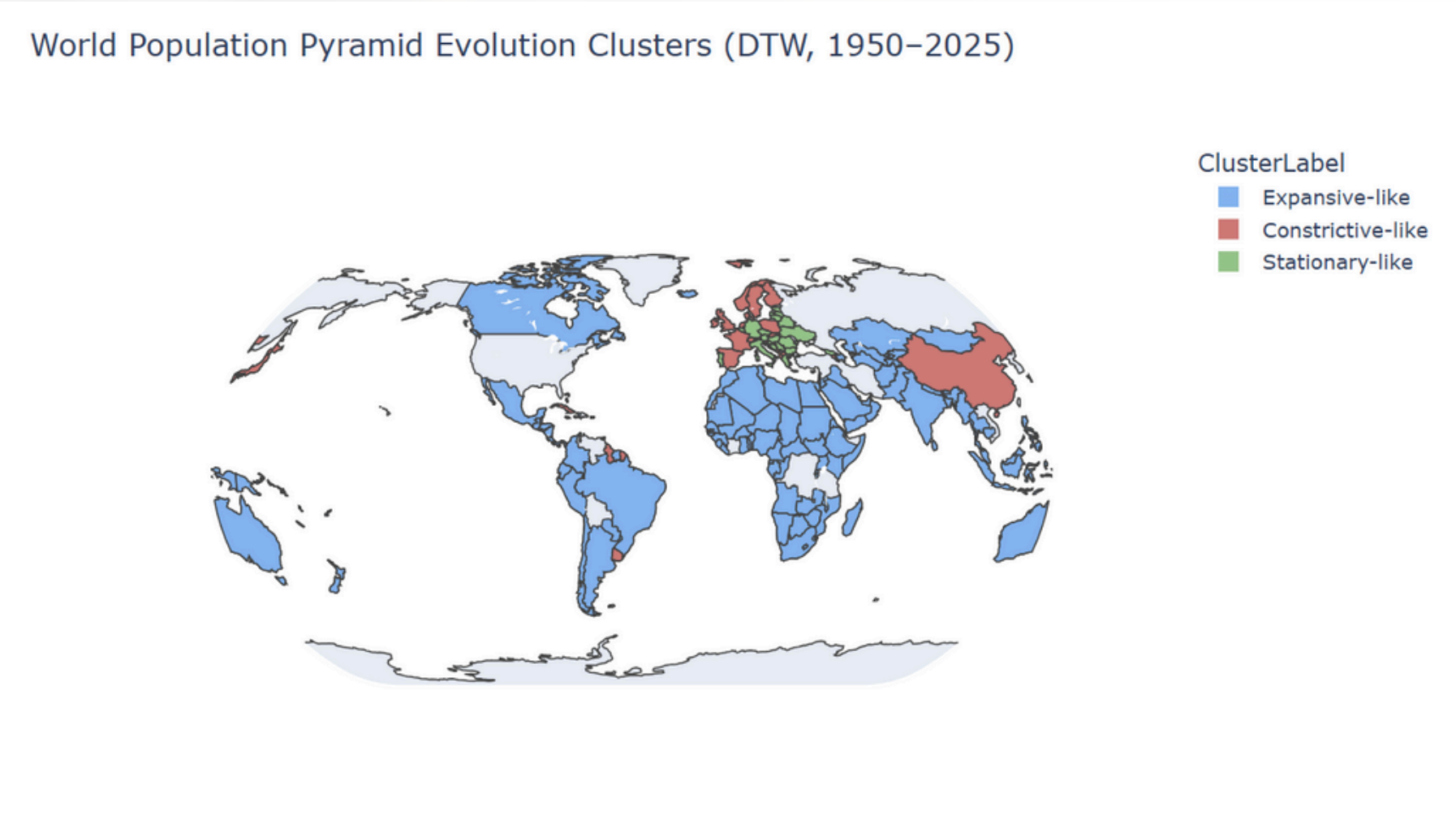
ClusterType	Countries
Constrictive	Australia, Australia/New Zealand, Austria, Belarus, Belgium, Bulgaria, Canada, Croatia, Czechia, Denmark, Estonia, Finland, France, Georgia, Germany, Greece, Hungary, IPCC AR6: Australia, Japan, and New Zealand, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Puerto Rico, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Ukraine, United Kingdom, Uruguay
Expansive	Afghanistan, Africa, Algeria, American Samoa, Angola, Bahrain, Bangladesh, Belize, Benin, Bhutan, Bolivia (Plurinational State of), Botswana, Brazil, Brunei Darussalam, Burkina Faso, Burundi, Cambodia, Cameroon, Chad, Colombia, Comoros, Congo, Costa Rica, Djibouti, Ecuador, Egypt, Equatorial Guinea, Eritrea, Eswatini, Ethiopia, Gabon, Gambia, Ghana, Guatemala, Guinea, Haiti, Honduras, India, Indonesia, Iran (Islamic Republic of), Iraq, Jordan, Kenya, Kiribati, Kuwait, Lebanon, Lesotho, Liberia, Libya, Madagascar, Malawi, Malaysia, Maldives, Mali, Mauritania, Mexico, Mongolia, Morocco, Mozambique, Myanmar, Namibia, Nepal, Nicaragua, Niger, Oman, Pakistan, Panama, Paraguay, Peru, Philippines, Qatar, Rwanda, Sao Tome and Principe, Saudi Arabia, Senegal, Sierra Leone, Singapore, Solomon Islands, South Sudan, Syrian Arab Republic, Tajikistan, Timor-Leste, Togo, Tunisia, Turkmenistan, Uganda, United Arab Emirates, United Republic of Tanzania, Uzbekistan, Vanuatu, Venezuela (Bolivarian Republic of), Yemen, Zambia, Zimbabwe
Stationary	Albania, Antigua and Barbuda, Argentina, Armenia, Azerbaijan, Bahamas, Barbados, Bosnia and Herzegovina, Chile, China, China, Taiwan Province of China, Cuba, Cyprus, Dominica, ESCAP: ENEA including Russian Fed (ENEA_RUS), El Salvador, Fiji, Grenada, Guam, Guyana, Israel, Jamaica, Kazakhstan, Kyrgyzstan, Mauritius, Montenegro, North Macedonia, Republic of Moldova, Saint Lucia, Saint Vincent and the Grenadines, Seychelles, Sri Lanka, Suriname, Thailand, Tonga, Trinidad and Tobago, United States Virgin Islands

Show 10 per page

RESULTS

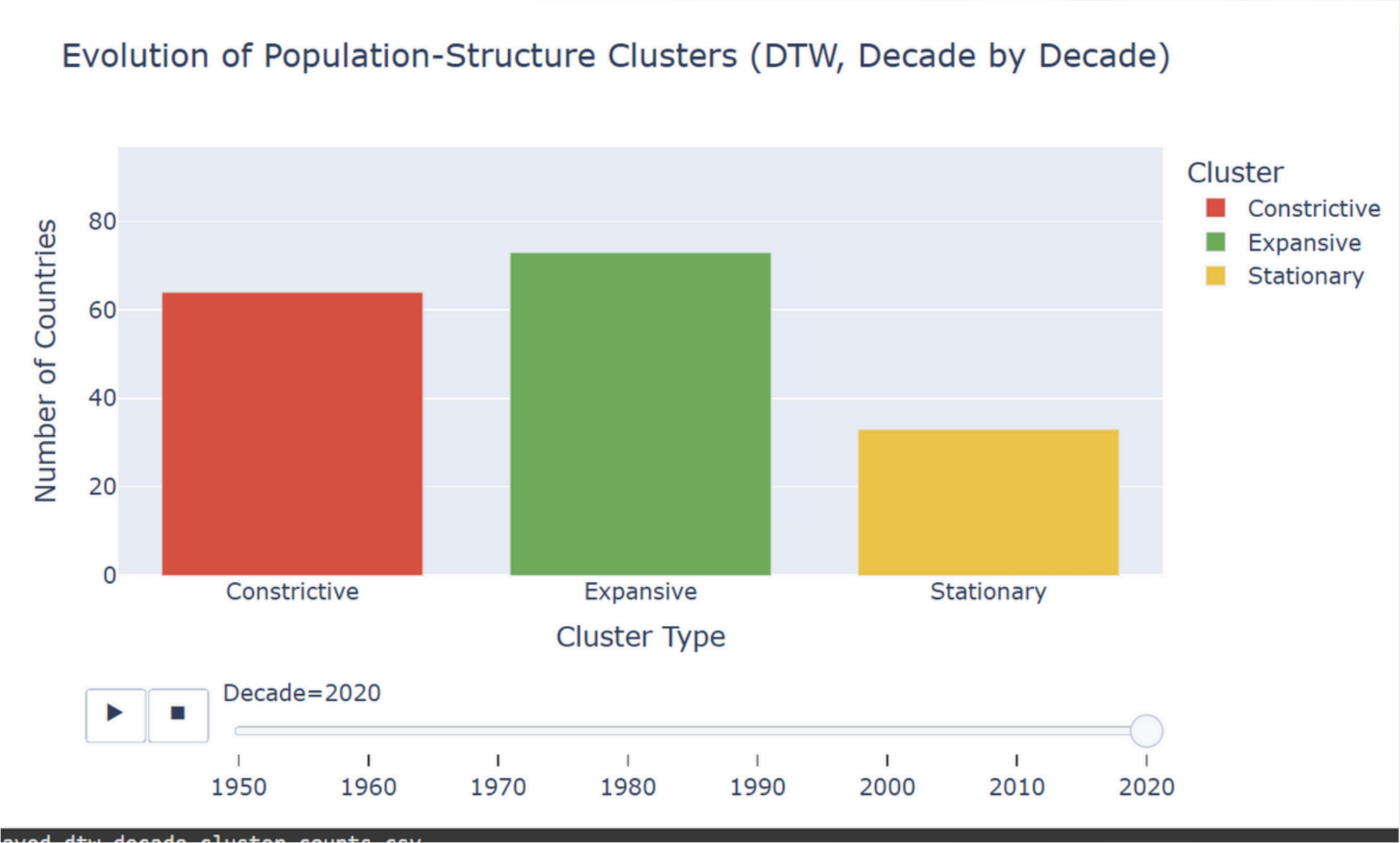
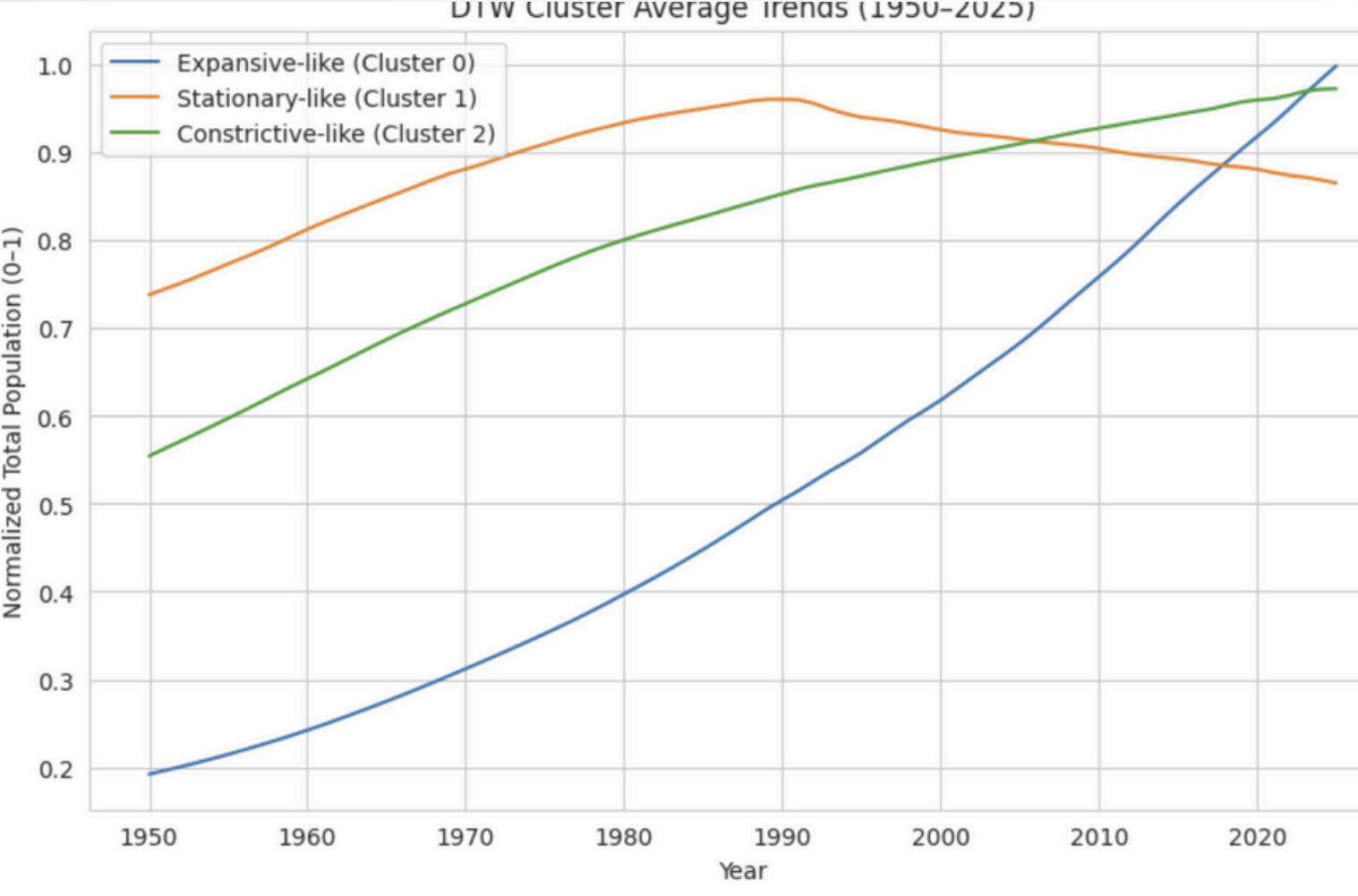


DTW CLUSTERING RESULTS



Armenia	0
Australia	0
Australia/New Zealand	0
Austria	0
Azerbaijan	2
Bahamas	0
Bahrain	2
Bangladesh	2
Barbados	0
Belarus	0
Belgium	0
Belize	2
Benin	1
Bhutan	2
Bolivia (Plurinational State of)	2
Bosnia and Herzegovina	0
Botswana	2
Brazil	0
Brunei Darussalam	2
Bulgaria	0
Burkina Faso	1
Burundi	1
Cambodia	2
Cameroon	1
Canada	0
Chad	1
Chile	0

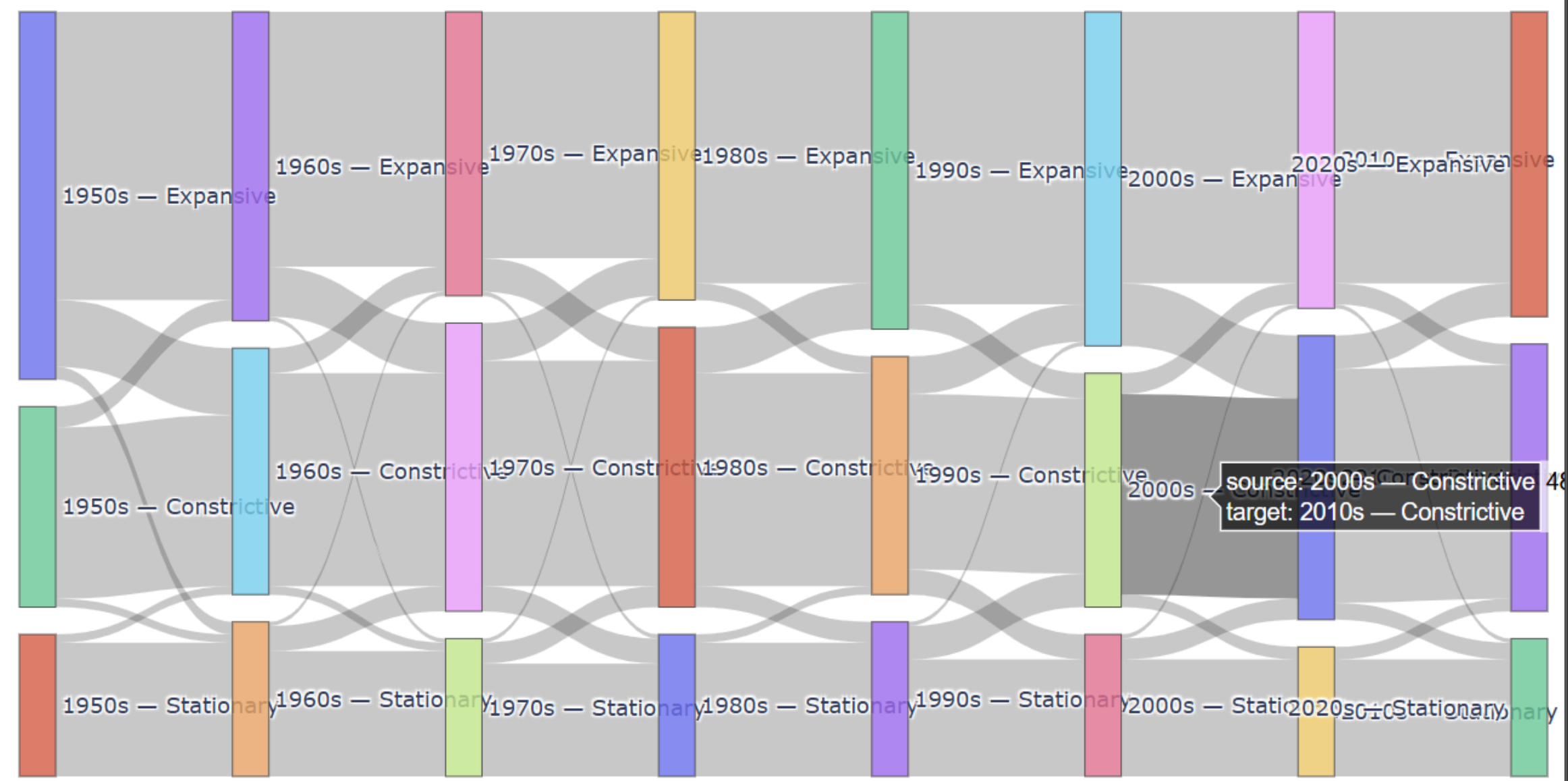
DTW CLUSTERING RESULTS



aved_dtw_decade_cluster_counts.csv

DTW CLUSTERING RESULTS

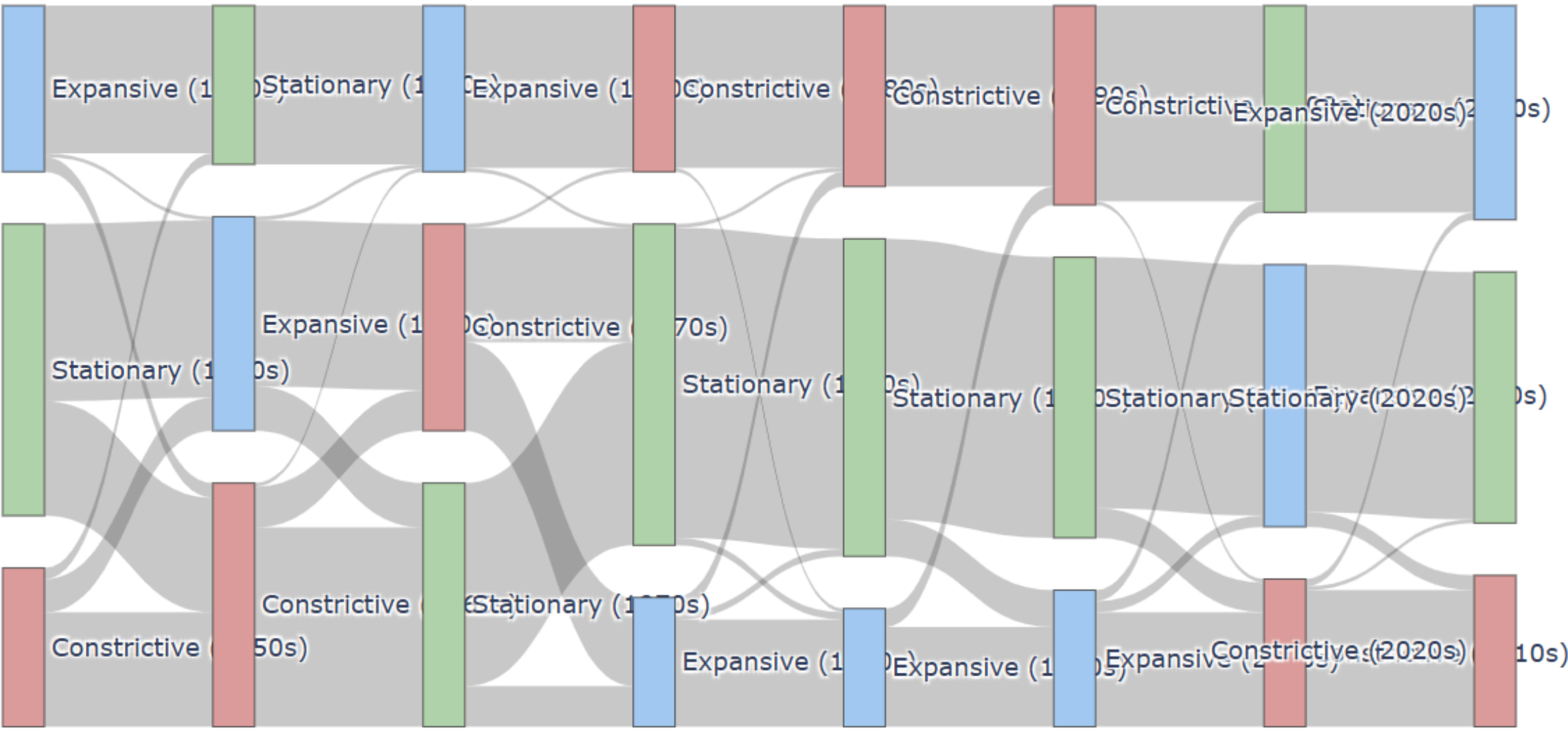
Sankey: DTW-derived cluster transitions by decade



TRANSITION DYNAMICS VISUALIZATION

-

Spectral Cluster Transitions (1950s–2020s)

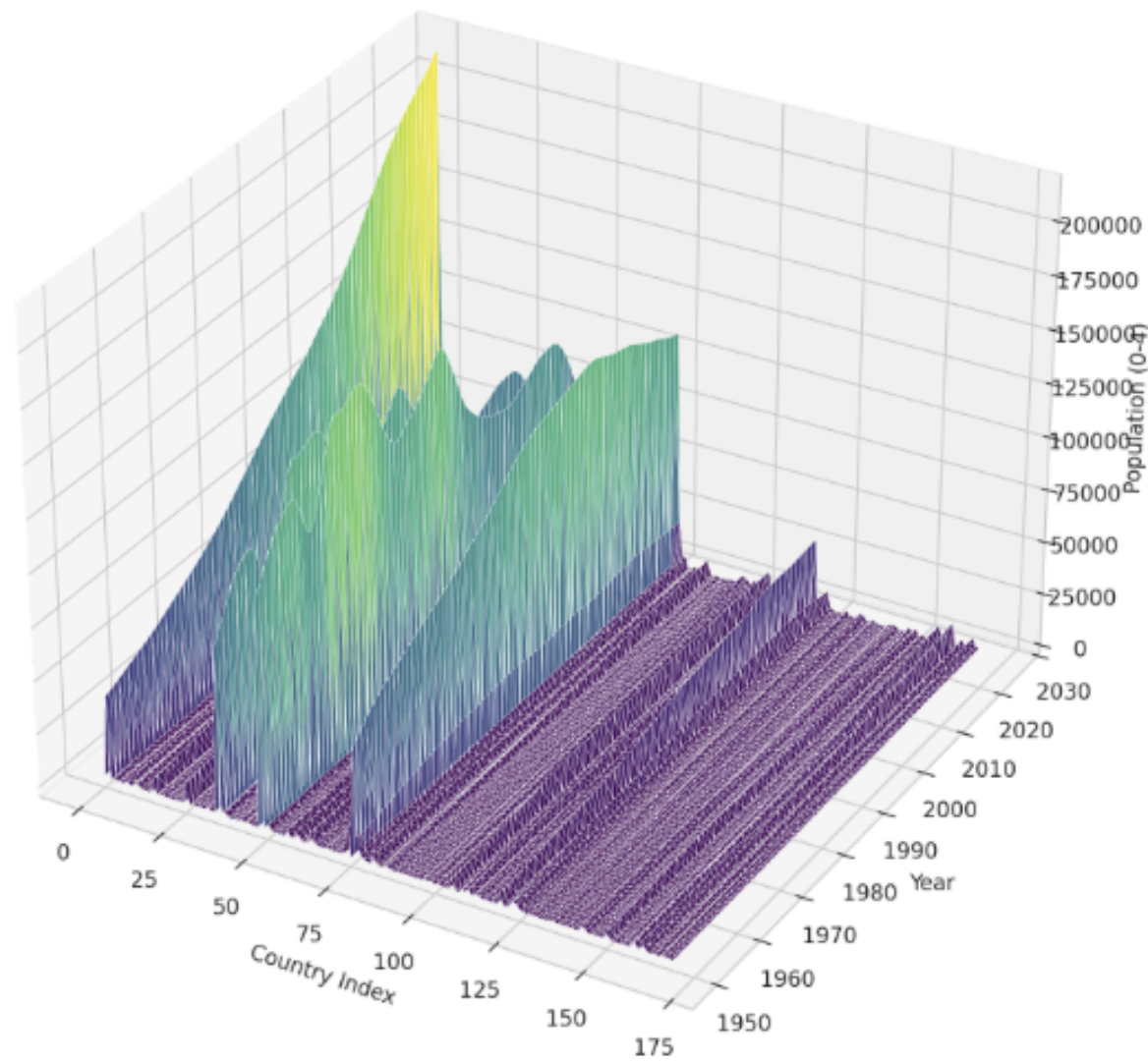


FUTURE WORK

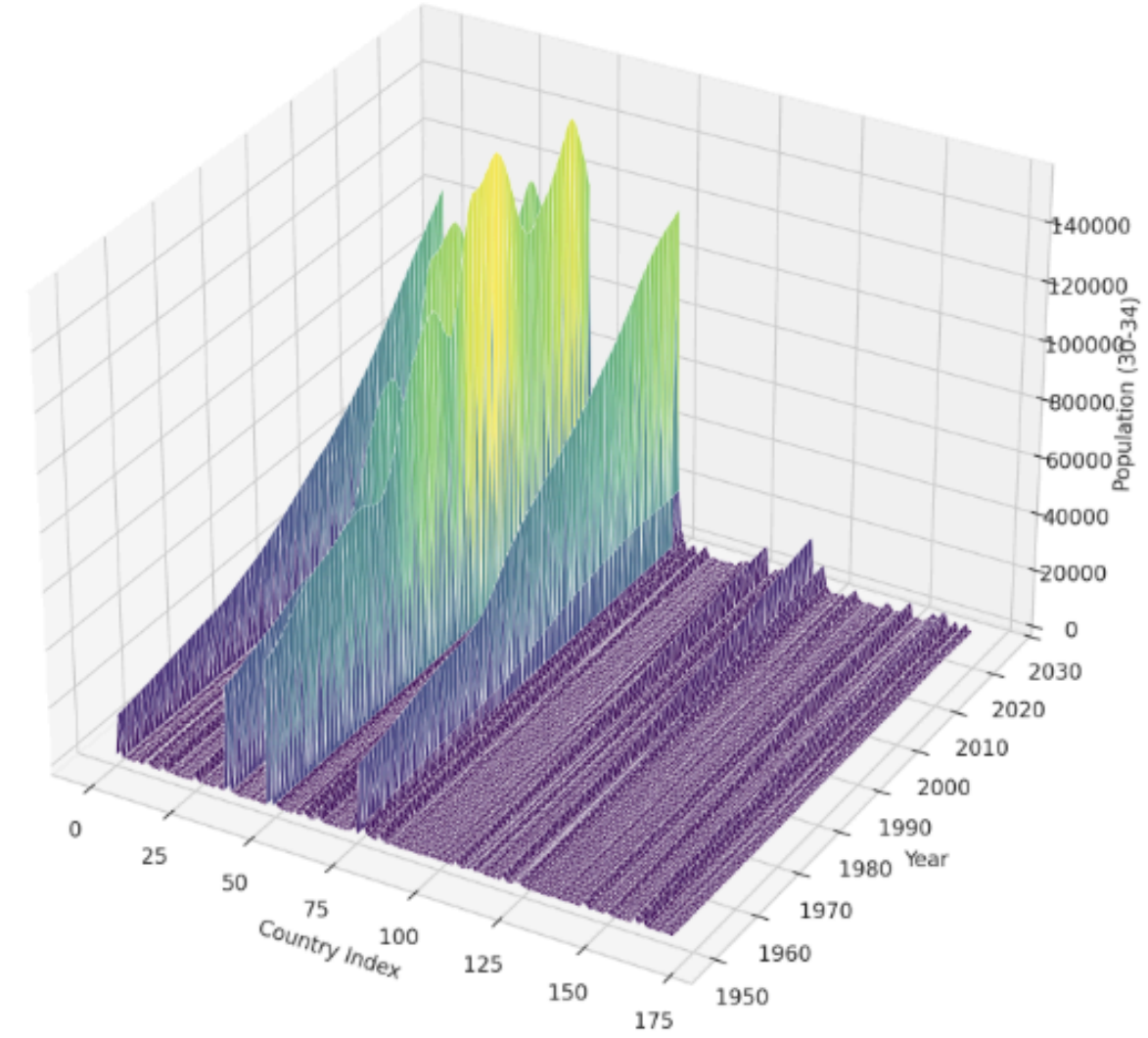
- **INCLUDE FORECAST TO 2050 (UN PROJECTIONS).**
- **ADD ECONOMIC & SOCIAL INDICATORS TO CLUSTERS (FERTILITY, GDP).**
- **EXPLORE DEEP LEARNING TEMPORAL EMBEDDINGS FOR PYRAMID EVOLUTION.**

DEMOGRAPHIC DIVIDEND ANALYSIS

3D Population Time Series — Age Group 0-4



3D Population Time Series — Age Group 30-34



Clustering of Population Pyramids (Korenjak-Černe et al., 2008)

- The study applied hierarchical clustering to group world countries by the shapes of their population pyramids over different years.
- It analyzed age-sex distributions of populations and identified three main pyramid types: Expansive, Stationary, and Constrictive, which represent different demographic transition stages.
- Used Ward's hierarchical clustering on data from 215 countries (1996–2006) and also applied the same method to 3,111 US counties.
- Found that most countries remained stable in their cluster category over time, but some showed transitions between demographic stages.
- Cluster stability was strong – especially in developed regions – showing long-term demographic consistency.
- Results highlight that local and regional factors (like education or migration) strongly influence pyramid shapes and demographic patterns.



RESEARCH BACKGROUND

Which country epitomizes the world? A study from the perspective of demographic composition — Yoshida et al. (2018)

- The study examines which country best represents the world's population structure over time (1990 → 2080) using composition ratios of age groups.
- Uses Compositional Data Analysis (CoDA) with Aitchison distance to measure similarity between countries' population structures and the world's structure.
- Analyzes 233 countries (focus on 201 with population > 90,000) and clusters them based on their compositional similarity in 2015 and across time.
- Identifies different “epitomes of the world” in different eras:
 - • 1990s → India, Algeria, Egypt
 - • 2015–2030 → Latin America (Argentina etc.)
 - • 2040 → New Zealand, U.S., Australia
 - • 2050–2060 → Uruguay, Puerto Rico
 - • 2080 → Japan, Italy (aging populations)
- Finds seven clusters in population structure, ranging from youthful pyramids (Sub-Saharan, Southern Asia) to aging “coffin-shaped” forms (Japan, Europe).
- Concludes that focusing only on total population or single indicators is insufficient — it's important to analyze the entire compositional structure across ages and time.



RESEARCH BACKGROUND

PDF: <https://arxiv.org/pdf/1810.00210>

A top-down view of a desk with various items: a laptop in the upper center, a cup of coffee on the left, a pen and glasses in the middle, paper clips on the left, and a large leaf in the bottom left corner. The background is a light, neutral color.

THANK YOU